

Xiaocong Yang

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Education

Incoming Ph.D. in Computer Science University of Illinois Urbana-Champaign	2025.8 - 2029.5 (expected)
M.S. in Computer Science (research-oriented) University of Illinois Urbana-Champaign	2023.8 - 2025.5
Courses: Efficiency in NLP (A+), Understanding LLMs (A), Optimization in Computer Vision (A)	
B.S. in Economics Tsinghua University	2018.9 - 2023.6
Courses: Data Structures and Algorithms (A+), Java Programming (A+), Probability and Stochastic Process (A+), Deep Learning (A), Database (A), Probability and Mathematical Statistics (A), Multivariate Calculus (A), Linear Algebra (A), Social Statistics (A)	

Publication

- Xiaocong Yang**, Jiacheng Lin, Ziqi Wang, Chengxiang Zhai. *Learning by Analogy: Enhancing Few-Shot Prompting for Math Word Problem Solving with Computational Graph-Based Retrieval*, in **submission to ACL 2025** [[link](#)]
- Ziyi Chen, **Xiaocong Yang**, Jiacheng Lin, Chenkai Sun, Jie Huang, Kevin Chen-Chuan Chang. *Cascade Speculative Drafting for Even Faster LLM Inference*, **NeurIPS 2024** [[link](#)]
- Xiaocong Yang**, James Huang, Wenxuan Zhou, Muhao Chen. *Parameter-efficient Tuning with Special Token Adaptation*, **EACL 2023** [[link](#)]
- Xingcheng Yao, Yanan Zheng, **Xiaocong Yang**, Zhilin Yang. *NLP from Scratch Without Large-Scale Pretraining: A Simple and Efficient Framework*, **ICML 2022** [[link](#)]
- Leixian Shen, Enya Shen, Yuyu Luo, **Xiaocong Yang**, Xuming Hu, Xiongshuai Zhang, Zhiwei Tai, Jianmin Wang. *Towards natural language interfaces for data visualization: A survey*, **TVCG 2022** [[link](#)]
- Hao Zhou, Pei Ke, Zheng Zhang, Yuxian Gu, Yinhe Zheng, Chujie Zheng, Yida Wang, Chen Henry Wu, Hao Sun, **Xiaocong Yang**, Bosi Wen, Xiaoyan Zhu, Minlie Huang, Jie Tang. *Eva: An open-domain chinese dialogue system with large-scale generative pre-training*, preprint [[link](#)]

Research Experience

Student Researcher TIMAN Group, University of Illinois Urbana-Champaign	2024.3 - now
Advisor: Chengxiang Zhai Donald Biggar Willett Professor, UIUC	
Urbana-Champaign	
- Proposed and led research on <i>Learning by Analogy</i>, a RAG system in which the retriever is trained to retrieve examples with similar computational graphs to enhance few-shot prompting by providing the LLM with the correct reasoning path. Our approach outperforms the baseline by 6.7 percent in absolute value. - Paper is in submission to ACL 2025.	
Student Researcher FORWARD Data Lab, University of Illinois Urbana-Champaign	2023.8 - 2024.1
Advisor: Kevin Chenchuan Chang Professor, UIUC	
Urbana-Champaign	
- Proposed <i>Cascade Speculative Drafting</i>, where we vertically and horizontally cascade multiple draft models of different size. Our approach is mathematically proved to be more efficient than standard speculative decoding and achieves up to 72 percent additional speedup in the experiments. - Paper is published at NeurIPS 2024.	
Visiting Researcher LUKA Group, University of Southern California	2022.4 - 2022.9
Advisor: Muhao Chen Assistant Professor, Information Science Institute, USC	
Los Angeles	
- Proposed and led research on <i>PASTA</i>, a model finetuning method that is more efficient than all of existing baselines which tunes only 0.029% of total parameters to match the performance of full finetuning. - Illustrated special tokens' role as information disseminators in LLMs. - Paper is published at EACL 2023.	
Student Researcher IIS, Tsinghua University	2021.7 - 2023.1
Advisor: Zhilin Yang Assistant Professor, Institute for Interdisciplinary Information Sciences, Tsinghua University Founder & CEO, Moonshot AI	
Beijing	
- Contributed to <i>Task-driven Language Modeling</i>, proposing a task-specific pretraining framework that reduced by 100x the total training cost and size of corpus and democratizing the development of large LLMs. - Analyzed the functionality of "vertical attention heads" in LLMs. - Paper is published at ICML 2022.	

Student Researcher | Conversational AI (CoAI) Group

2021.1 - 2021.6

Advisor: [Huang Minlie](#) | Associate Professor | Department of Computer Science and Technology, Tsinghua University
Beijing

- Contributed to the development of the open-domain dialogue system *EVA* and the corresponding paper.
- Designed the interface and client-server communication framework for the deployment of the dialog system, tested rule-based and model-based methods to filter out toxic content in the generated text.

Teaching Experience

- ♦ **Teaching Assistant** | CS107 Data Science Discovery, UIUC 2023.8 - 2023.12
- ♦ **Teaching Assistant** | CS441 Applied Machine Learning, UIUC 2024.1 - now

Skills

- **Programming:** Python, Java, C, C++, Bash
- **Software Libraries:** Pytorch, Transformers, Scikit-learn, Seaborn
- **Research Interests:** Machine learning in large language models, with a focus on efficiency, interpretability and fairness

Diversity and Inclusion

- Student Global Member | **oSTEM (Out in Science, Technology, Engineering, and Mathematics)** 2022.11 - now
- Student Member | **oSTEM @ Illinois** 2023.11 - now